EXPLORATORY DATA ANALYSIS USING POWER BI

WEEK 7 – STUDENT PERFORMANCE

DATA CLEANING & TRANSFORMATION

Submitted by :

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1. INTRODUCTION

This focuses on cleaning, transforming, and preparing the Student Performance dataset using **Power BI**. The raw data was examined for errors, duplicates, missing values, and inconsistent data. New calculated columns and measures were created to analyze students' performance effectively.

2. OBJECTIVES

- To import the Student Performance dataset into Power BI.
- To inspect and clean the dataset.
- To identify and handle duplicates, missing values, and errors.
- To standardize categorical values and data types.
- To create Total Score and Average Score columns.
- To classify students based on Performance Grade.
- To create Pass/Fail classification.
- To unpivot subject score columns.
- To create DAX measures for performance analysis.

3. TOOLS USED

- **Power BI Desktop**
- **Power Query Editor**
- **DAX (Data Analysis Expressions)**
- Student Performance in Exams Dataset

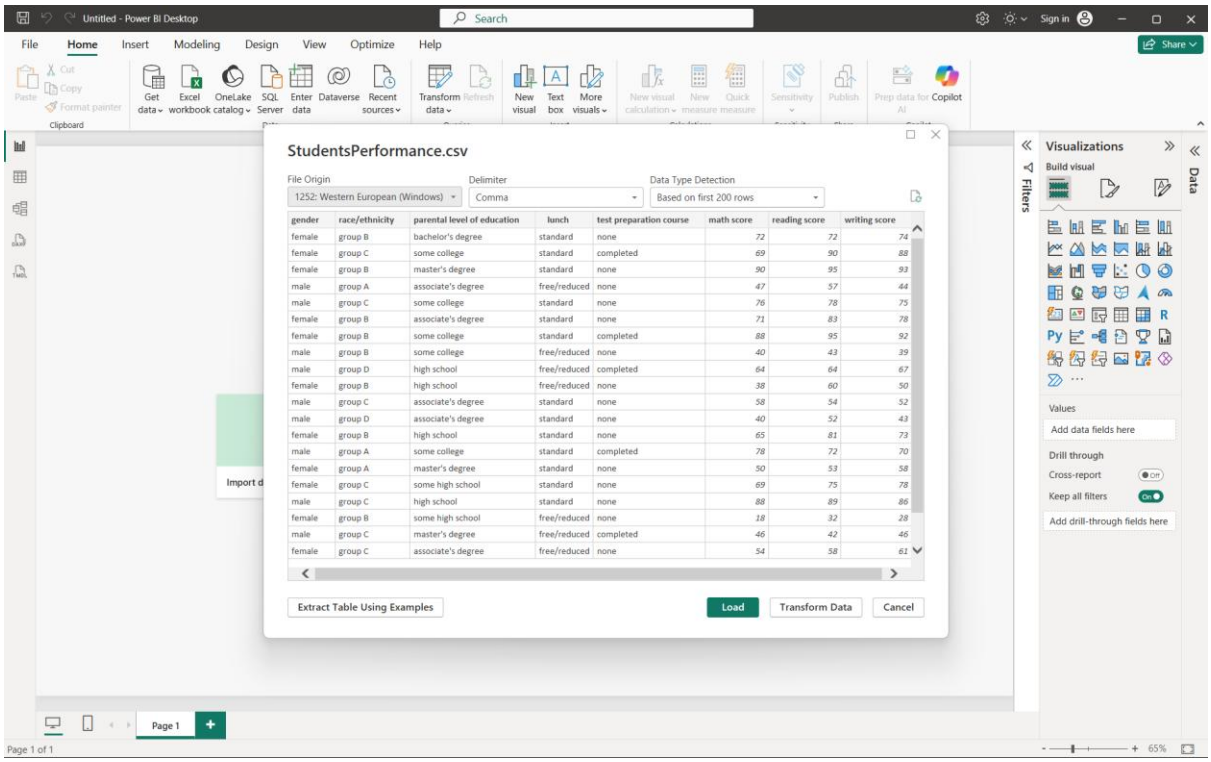
4. IMPORTING THE DATASET

The Student Performance dataset was imported into Power BI using the **Get Data** option. The data was loaded into Power Query Editor for further inspection and transformation.

Procedure:

1. Open Power BI Desktop.
2. Select **Home → Get Data**.
3. Select the appropriate file format.
4. Choose the Student Performance dataset.
5. Select **Transform Data** to open Power Query Editor.

Dataset Imported into Power BI:



5. DATASET INSPECTION

The dataset was examined to understand its rows, columns, data types, and different categorical and numerical attributes.

Numerical Attributes

- Math Score
- Reading Score
- Writing Score

Categorical Attributes

- Gender
- Race/Ethnicity
- Parental Education
- Lunch
- Test Preparation Course

The data types of the columns were checked and corrected where required.

6. DATA CLEANING

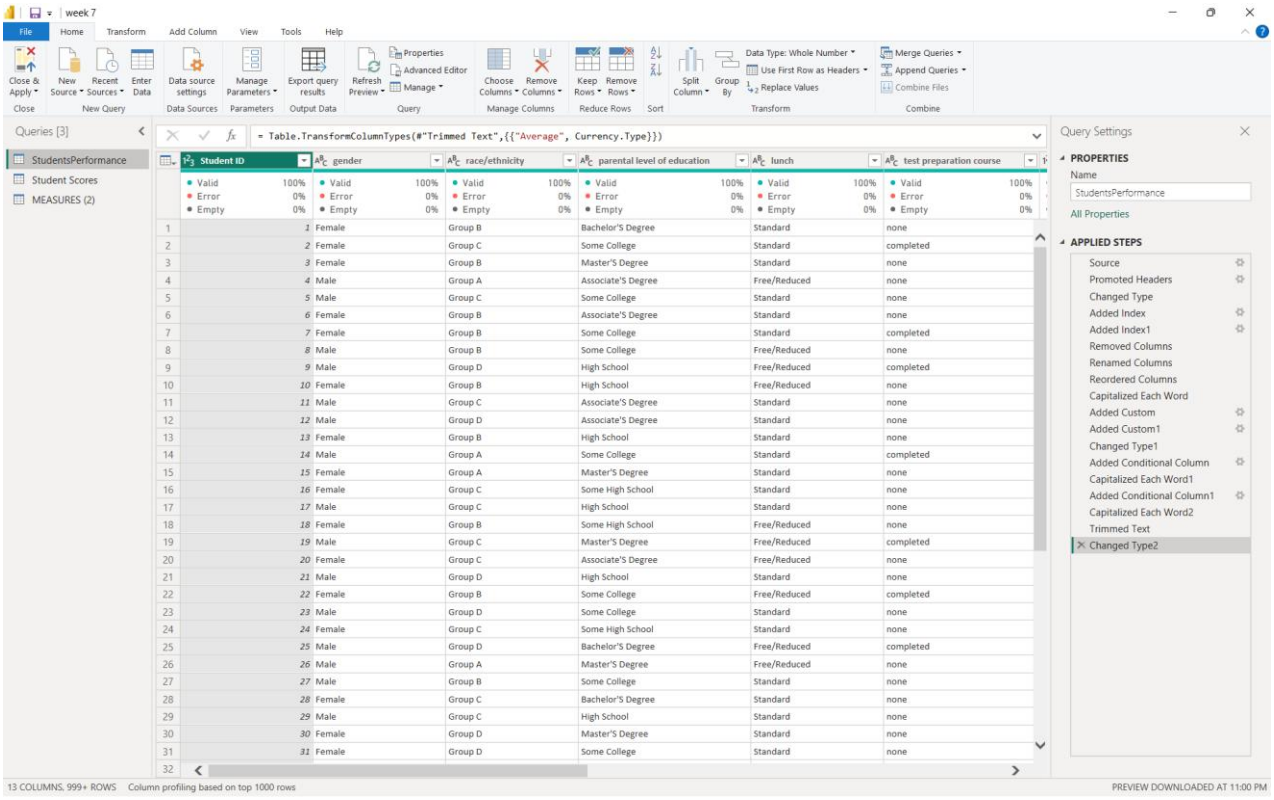
The dataset was cleaned using Power Query Editor.

The following operations were performed:

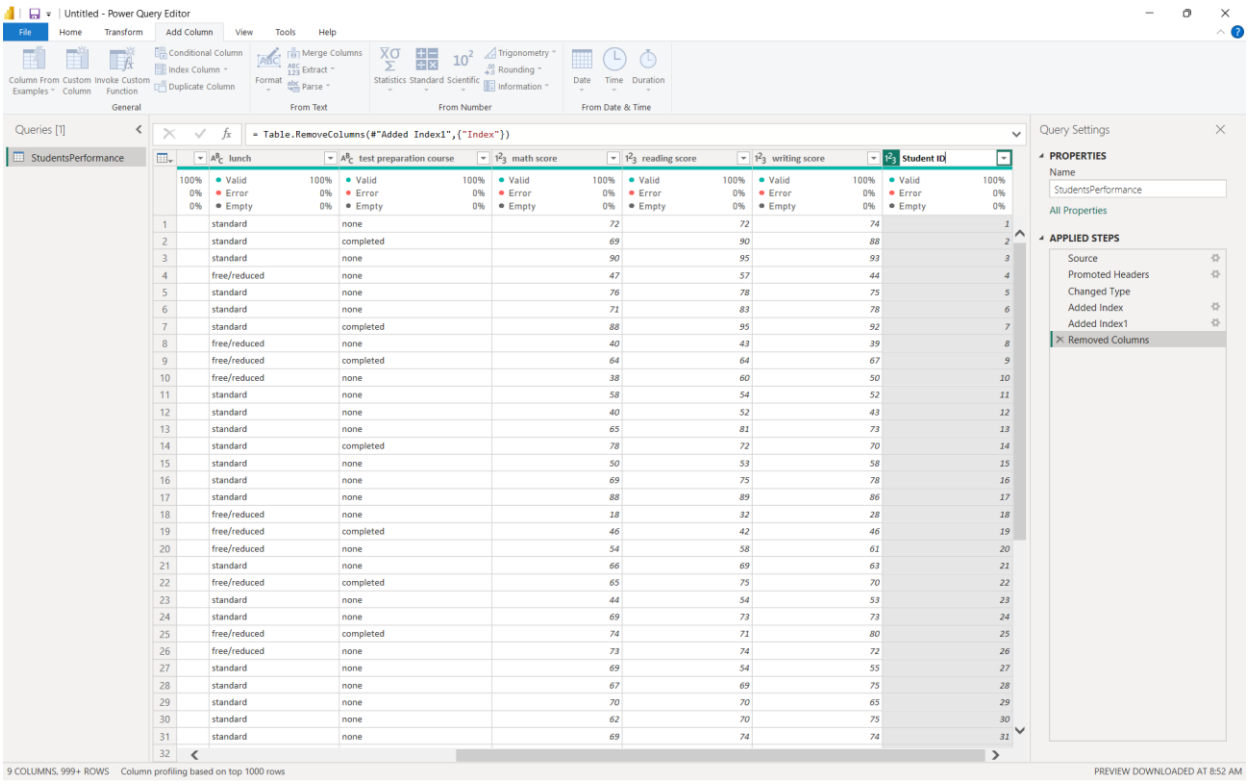
- Checked for duplicate rows.
- Checked for missing or null values.
- Checked for errors.
- Verified appropriate data types.
- Removed unnecessary spaces.
- Standardized categorical values.
- Renamed columns where necessary.

The cleaned dataset was then reviewed to ensure that the data was consistent and ready for transformation.

Data Cleaning in Power Query:



Student ID were also created using Index column(from index 1):



7. CREATING TOTAL MARKS

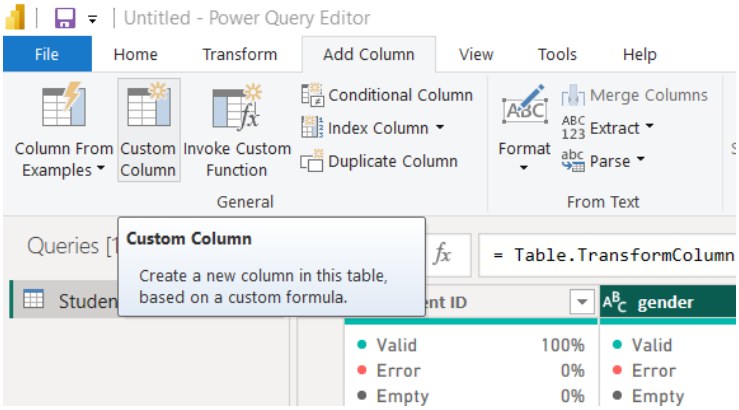
A **Total Marks** column was created by adding the Math, Reading, and Writing scores.

Power Query Formula

Total Marks = [Math Score] + [Reading Score] + [Writing Score]

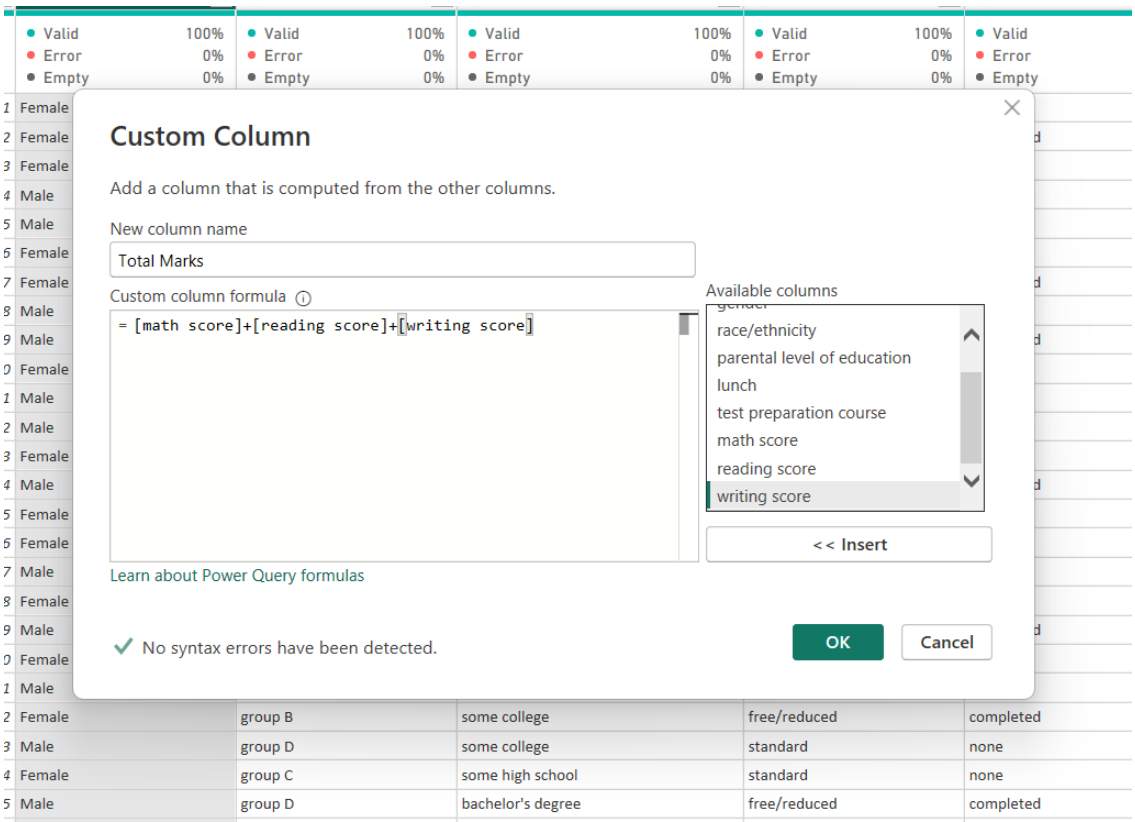
Procedure:

1. Go to **Add Column**.
2. Select **Custom Column**.



3. Enter the formula.
4. Name the column **Total Marks**.
5. Click **OK**.

Total Marks Column:



8. CREATING AVERAGE SCORE

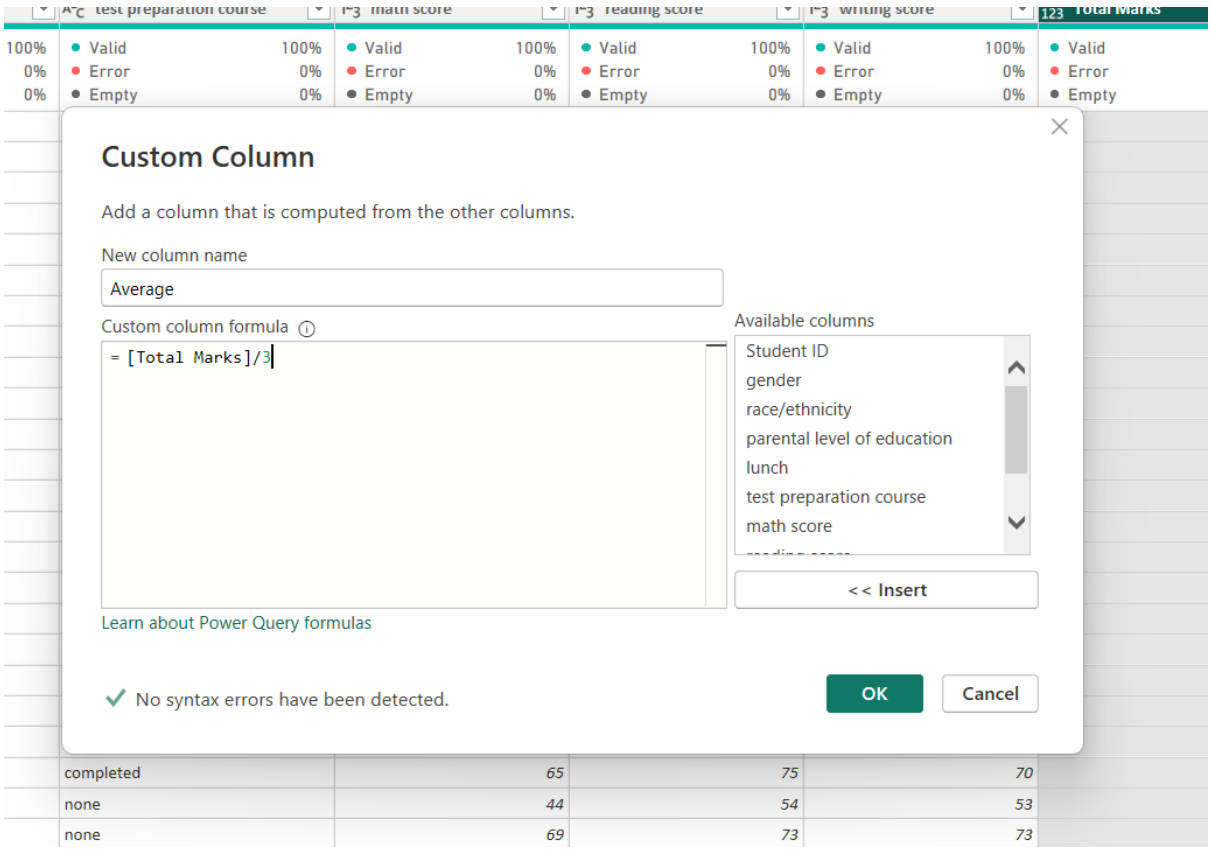
An **Average Score** column was created to calculate the student's average performance across the three subjects.

Formula

Average Score = [Total Score] / 3

This gives a single score representing the student's overall academic performance.

Average Score Column:



9. CREATING PERFORMANCE GRADE:

A **Grade** column was created using conditional logic based on the Average Score.

The following classification was used:

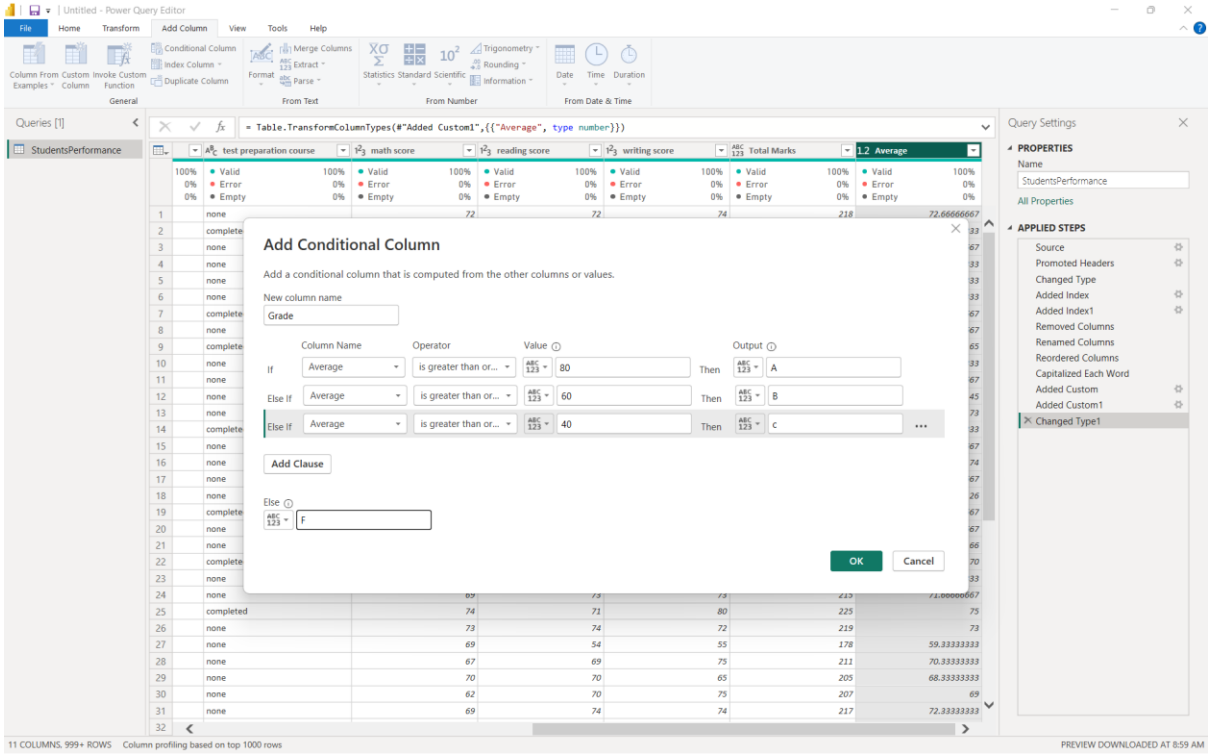
Average Score	**Grade**
80 – 100	A
60 – 79	B
40 – 59	C
Below 40	F

Procedure

1. Select **Add Column**.
2. Select **Conditional Column**.
3. Select **Average**.

4. Add the required conditions.

5. Name the column **Grade**.



10. CREATING RESULT COLUMN

A **Result** column was created based on the student's Average Score.

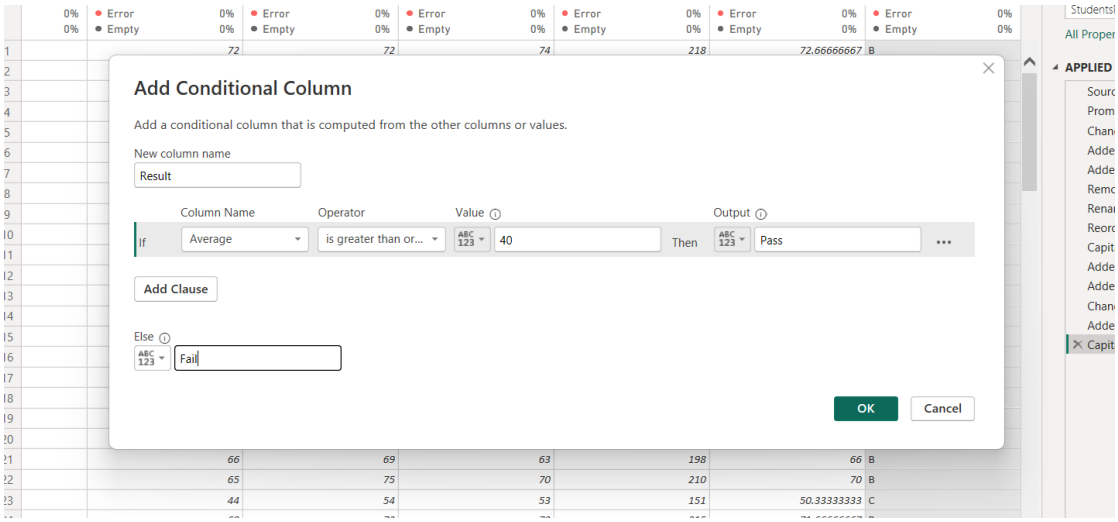
The criterion used was:

- **Average Score ≥ 40 → Pass**
- **Average Score < 40 → Fail**

Procedure

1. Select **Add Column → Conditional Column**.
2. Select **Average Score**.
3. Set the condition as greater than or equal to 40.
4. Enter **Pass** for the true condition.
5. Enter **Fail** for the false condition.

Pass/Fail Classification:



Therefore, those four columns were created :

Total Marks	Average	Grade	Result
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
218	72.67	B	Pass
247	82.33	A	Pass
278	92.67	A	Pass
148	49.33	C	Pass
229	76.33	B	Pass
232	77.33	B	Pass
275	91.67	A	Pass
122	40.67	C	Pass
195	65.00	B	Pass
148	49.33	C	Pass
164	54.67	C	Pass
135	45.00	C	Pass
219	73.00	B	Pass
220	73.33	B	Pass
161	53.67	C	Pass
222	74.00	B	Pass
263	87.67	A	Pass
78	26.00	F	Fail
134	44.67	C	Pass
173	57.67	C	Pass
198	66.00	B	Pass
210	70.00	B	Pass
151	50.33	C	Pass
215	71.67	B	Pass
225	75.00	B	Pass
219	73.00	B	Pass
178	59.33	C	Pass
211	70.33	B	Pass
205	68.33	B	Pass
207	69.00	B	Pass
217	72.33	B	Pass

11. UNPIVOTING SUBJECT SCORES

The three subject columns:

- Math Score
- Reading Score
- Writing Score

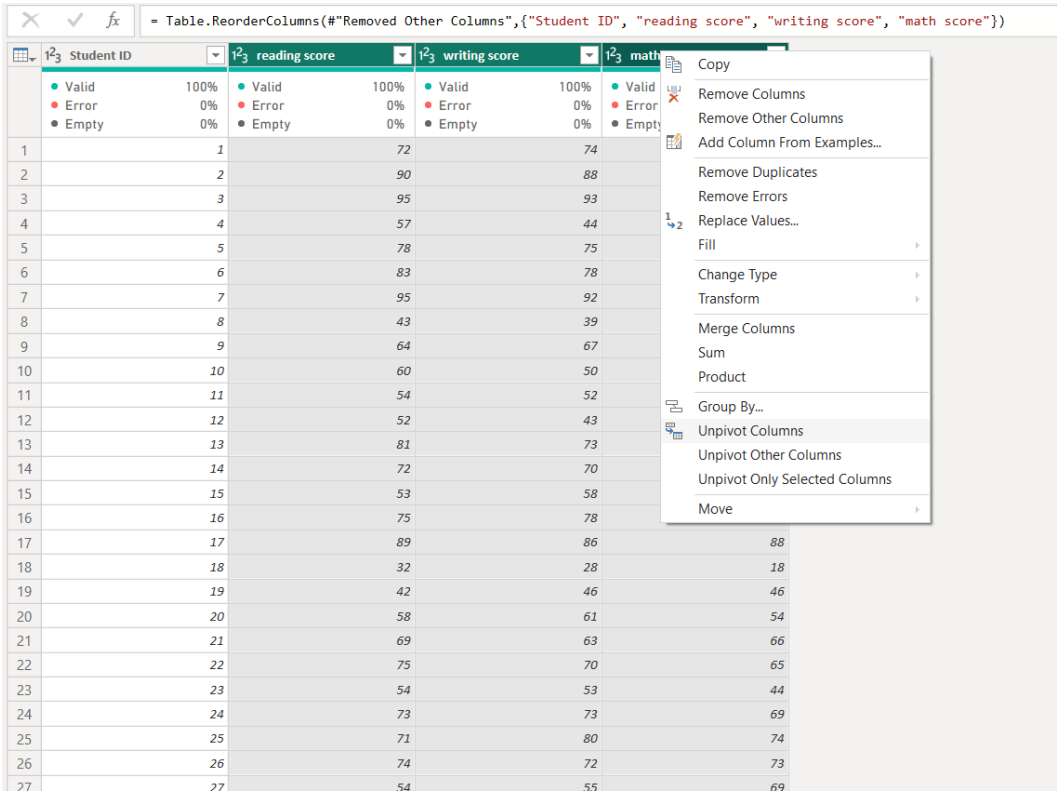
were transformed into a structured format using the **Unpivot Columns** option.

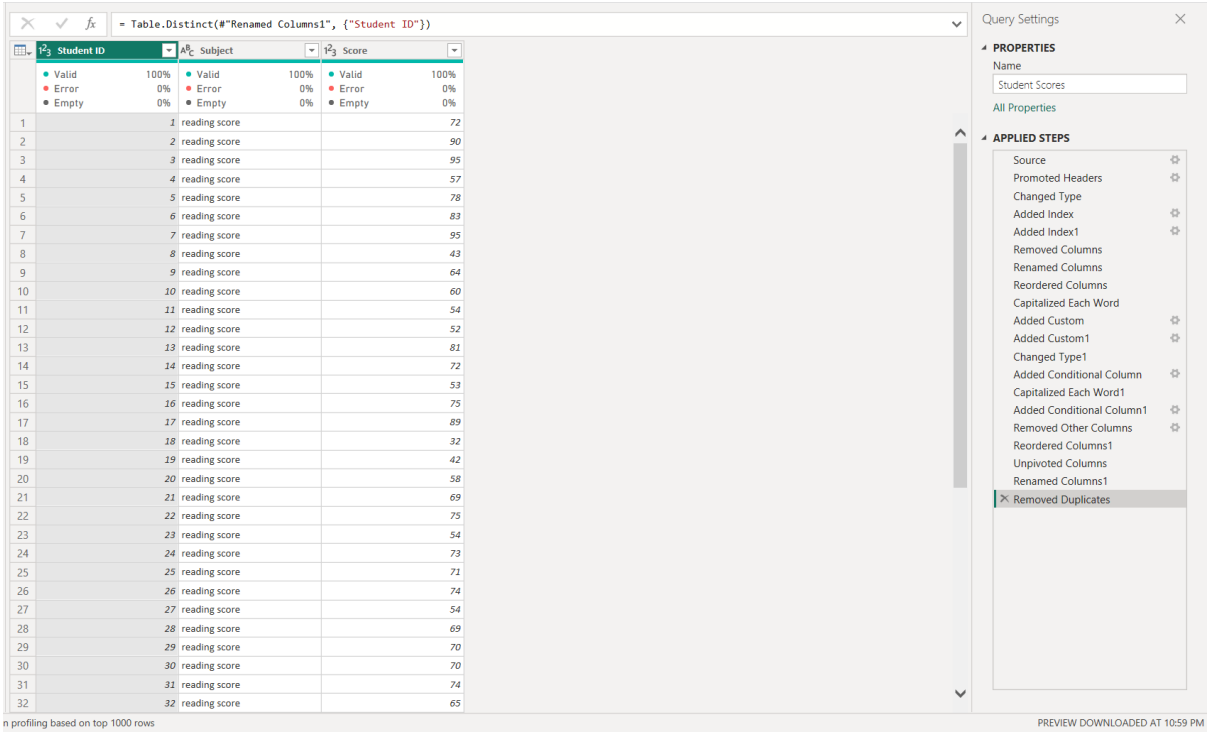
Procedure

1. Select the three subject score columns.
2. Right-click and select **Unpivot Columns**.
3. Rename **Attribute** as **Subject**.
4. Rename **Value** as **Score**.

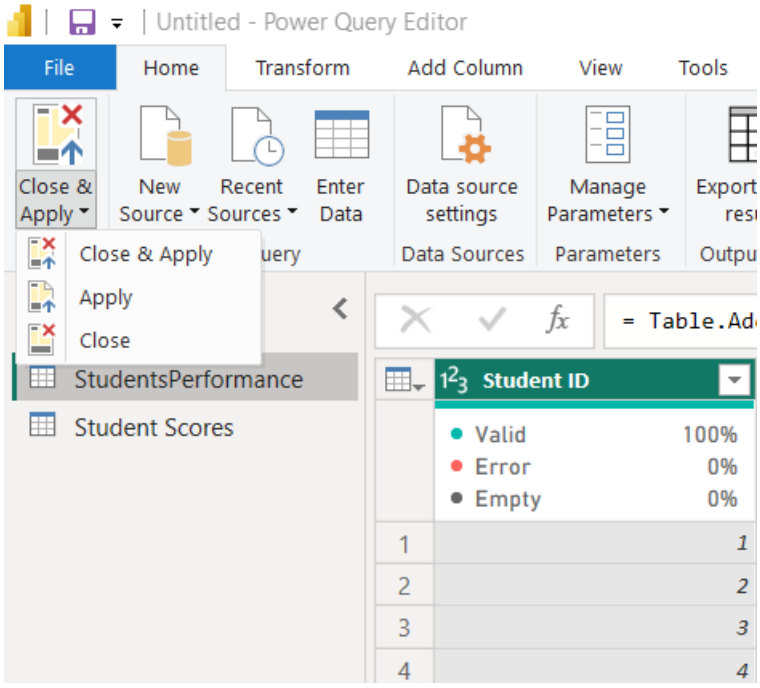
This transformation makes it easier to compare student performance across subjects in Power BI visuals.

Unpivoted Subject Data:





Then , close & apply.



12. PREPARED DATA MODEL

After completing the transformations, the data model was reviewed to ensure that the required columns were available for analysis.

The prepared dataset contains:

- Student demographic information
- Subject scores
- Total Marks
- Average
- Grade
- Result
- Subject
- Score

13. CREATING DAX MEASURES

After applying the Power Query transformations, DAX measures were created for performance analysis.

Average Math score:

```
Average Math score = AVERAGE('Student Performance'[Math Score])
```

Average Reading score:

```
Average Reading score = AVERAGE('Student Performance'[Reading Score])
```

Average Writing score:

```
Average Writing = AVERAGE('Student Performance'[Writing Score])
```

Overall Average Score

```
Overall Average Score =

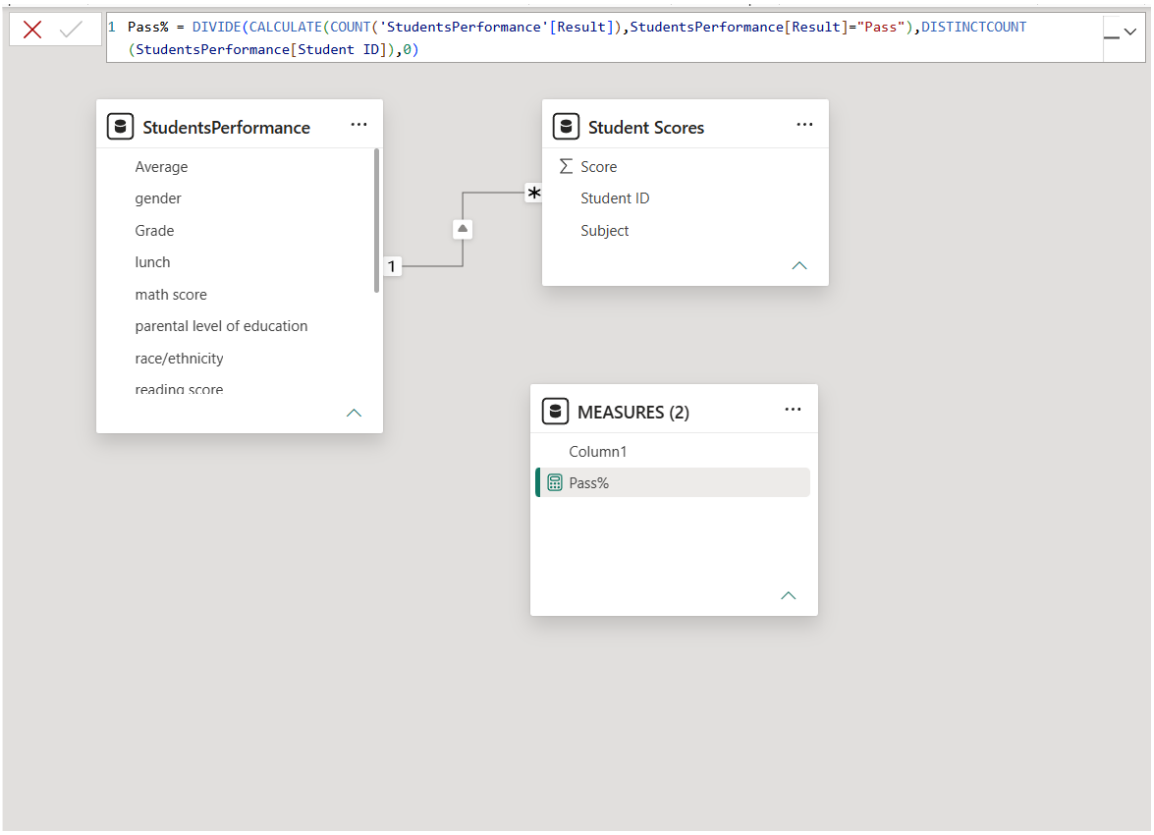
AVERAGE('Student Performance'[Average])
```

Overall Pass %

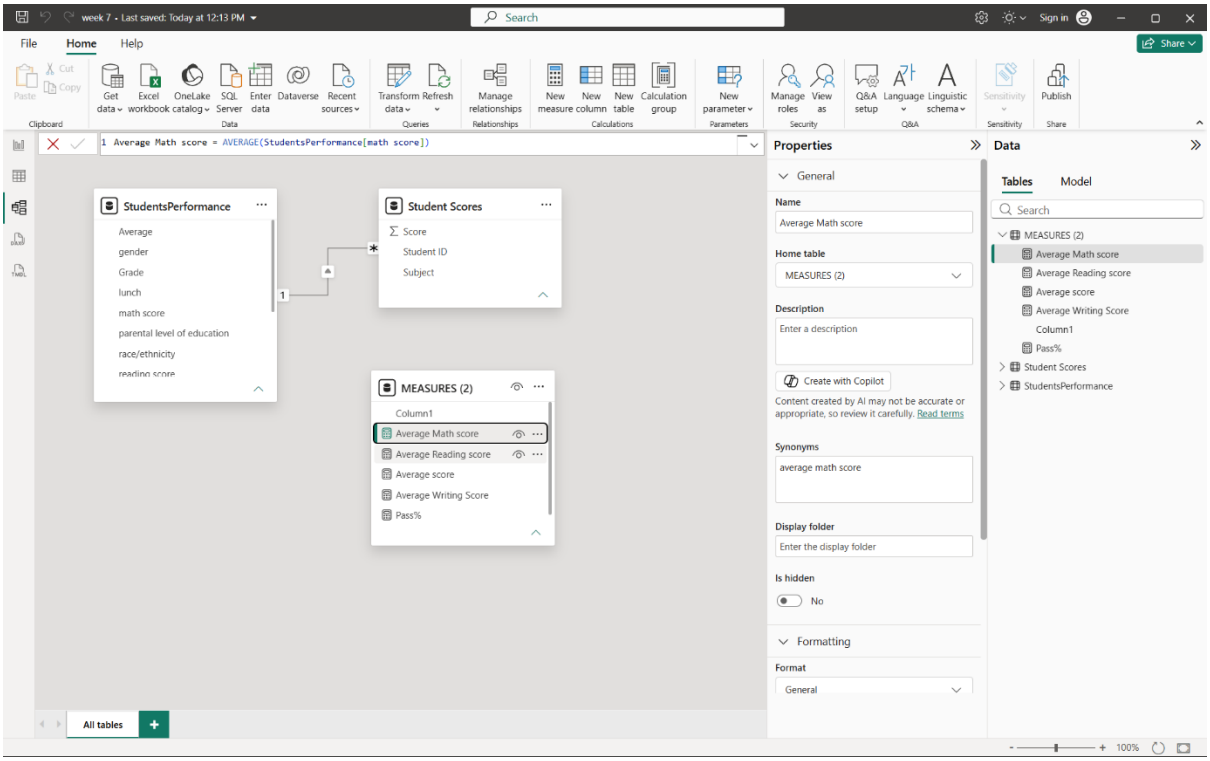
```
 Pass % =

DIVIDE(
    CALCULATE(
        COUNT('Student Performance'[Result]),
        'Student Performance'[Result] = "Pass"
    ),
    DISTINCTCOUNT('Student Performance'[Student ID]),0)
```

The **Overall Pass %** measure can be formatted as a percentage in Power BI.



Other DAX Measures:



14. VALIDATION

After completing the transformations and calculations, the dataset was validated.

The following checks were performed:

- Checked that there were no unwanted errors.
- Verified the calculated Total Score.
- Verified the Average Score.
- Checked the Performance Grade.
- Checked Pass/Fail classification.
- Verified the Subject and Score columns after unpivoting.
- Verified the DAX measures.
- Confirmed that the transformed data was ready for visualization.

15. RESULT

The Student Performance dataset was successfully cleaned and transformed using Power BI. New calculated columns such as **Total Marks, Average , Grade, and Result** were created. The subject scores were also unpivoted into **Subject** and **Score** columns.

The required DAX measures were created to support academic performance analysis. The prepared dataset can now be used to create an interactive **Student Performance Dashboard** in the next stage.

16. CONCLUSION

Week 7 activity provided practical experience in data cleaning, transformation, calculated columns, unpivoting, and DAX in Power BI. The final dataset is structured and ready for further analysis and dashboard development.